

INTRODUCTION TO HTML

- ⇒ HTML stands for Hypertext Markup Language.
- ⇒ HTML is used to create web page or web application.
- ⇒ We can create a static website by using HTML.
- ⇒ HTML is a markup language.

History of HTML:-

⇒ Tim Berners-Lee is the father of HTML.

Version of HTML:-

- HTML 1.0 - The first version of HTML was 1.0, which was the barebones version of HTML language and it is released in 1991.
- HTML 2.0 - This version was released in 1995 and it was standard language version for website design.
- HTML 3.2 - This version was released in 1997 and it was capable of creating tables and providing the extra support for form element.
- HTML 4.01 - This version was released in December 1999, and it is a very stable version of HTML language.
- HTML 5 - HTML 5 is the latest version of Hypertext Markup language.
 - The first draft of this version was announced in January 2008, and it was finally published in 2014.

Features of HTML:-

- ↳ It is easy to learn and easy to use.
- ↳ It is platform-independent.
- ↳ Image, video and audio can be added to a webpage.
- ↳ Hypertext can be added to the text.

What is HTML?

→ HTML stands for Hypertext Markup language.

→ It is the standard language used to create and design webpages can be added to the text on the internet.

Hypertext:-

⇒ Hypertext simply means "text within text", text has a link within it is a hypertext.

⇒ Hypertext is a way to link two or more webpage with each other.

Markup Languages:-

⇒ A ~~make~~ markup language is a computer language that is used to apply layout and formatting to a text document.

⇒ Markup language makes text more interactive & dynamic.

Webpage:-

⇒ A webpage is a document which is commonly written in HTML & translated by a web browser.

⇒ A webpage can be identified by entering an URL.

⇒ A webpage can be of static or dynamic type.

HTML Text Editor :-

An HTML file is a text file, so to create an HTML file we can use any text editor.

⇒ Text editors are the programs which allow editing in a written text, hence to create a webpage we need to write our code in some text editor.

e.g: Vscode, Notepad etc.

Basic HTML Document Structure :-

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="utf-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title> Document </title>

</head>

<body>

</body>

</html>

DOCTYPE :-

It defines the document type or it instructs the browser about the version of HTML.

HTML :- <html>

This tag describes the HTML document.

<Head> :- The head tag contains some meta tags.

<meta> :- Meta tag provides information to the current document.

<title> :- This tag defines the title name of HTML document & displays top of the window tag.

<body> :- Body tag is used for displaying the content on the webpage.

HTML Comments :-

It means a code written inside the HTML document but does not execute & can't display on webpage. That means add some note in HTML document.

<!-- <body>
 <div></div>

</body>-->

What is HTML ?

- * HTML is a markup language which is used for create webpage.
- * It is also called as skeleton of webpage.
- * A html file or document create a single webpage on the internet.
- * The html formatting the content & layout the structure by using different types of tags.

Page: It means writing something content & gain something information.

Webpage :-

- * It just like that written something content & display on the web or internet, This page is called web page.
- * This webpage is written content by using HTML.

Website :-

- * The collection of one or more webpage is called website.
- * A website can visit by using website domain name.

Web browser :-

- * Web browser is a software application that displaying the website such as Mozilla Firefox, Safari, Chrome etc.

Tag :-

- * Tag is a keyword i.e used for formatting the text & create layout in the webpage.
- * A tag define by using (<>) angular braces.
- * A tag can't display in the webpage.
- * It is case-sensitive.
- * There are two types of tags.
 - Pair Tag / Close
 - Unpair Tag / Empty Tag

Pair Tag:- A tag have starting tag and ending tag is called pair Tag.

Unpair Tag:- A tag have only starting tag.

Starting Tag:- It define the content will be start.

Ending Tag:- It define the content will be end.

HTML ELEMENT:-

* HTML element means starting tag, content & ending tag. That means all tags are elements. such as pair tag, unpair tag.

* There are three types of elements behaves different behaviour.

a) Block Element

b) Inline Element

c) Inline Block Element

a) Block Element Behaviour:-

↳ Block element takes total width of the webpage & height according to the content.

↳ It can used css height & width.

↳ It take margin, work horizontally & vertically.

padding also take both horizontally & vertically.

↳ The block element contain always start a new line

Such block elements names are : <div>, <h1>...<h6>, <heading>, <nav>
<p>, <div>, , , <dl>, <dt>, <dd>, <table>, <tr>, <form>

<pre>, <main>, <footer>... etc.

b) Inline Element Behaviour:-

↳ Inline element take height & width according to the content.

↳ It doesn't work css height & width property.

↳ It always start same line or ending content of inline element content.

↳ Margin will be work only horizontally. padding will be work

both horizontal & vertical but top & bottom padding overlap top & bottom elements.

Such inline elements name `<span>`, `<b>`, `<strong>`, `<mark>`, `<i>`, `<del>`... etc

c) Inline-Block Element:-

- ↳ It take height & width according to the content.
- ↳ It always start same line.
- ↳ It work CSS height & width property.
- ↳ Margin & padding will be work both horizontally & vertically.
- ↳ Such inline block elements: `<img>`, `<audio>`, `<video>`, `<button>`, `<input>`
`<textarea>`, `<select>`... etc.

Attribute:-

- ↳ In html, attribute provides some external infoⁿ about an element.

↳ Attribute can define by attribute name with value.

↳ Attribute name & value are case-sensitive.

↳ We can add more than one attribute in one element but give some space between attributes.

↳ Attribute always specified in the starting tag.

Syntax: `<element attribute1:"value", attribute2:"value"> content </element>`

↳ Ex of attribute: `html`, `lang`, `charset`, `name`, `content`

html Attribute: This attribute define in doc type element for describe which document.

Charset Attribute: This attribute define which type of character set used in document.

lang Attribute: It define which language used in document.

Name Attribute: It define the name of an element.

Content Attribute: It ~~define~~ contains the value for the name attribute depending on which is used.

title Attribute: It define title name to the html element.

style Attribute: It define style effect to the html element.

id Attribute: It define unique id to the html element for uniquely identify to the whole html doc.

Heading Tag:-

- * This tag is used for add main heading or sub heading content on the webpage.
- * It is a block element. It's default behaviour is text bold & add top & bottom margin.
- * There are six type. `<h1>` ---- `<h6>` to `<h2>` ---- `<h6>`

Syntax :- `<h1>` content `</h1>`

Paragraph Tag <p>:-

- * This tag display paragraph content on the webpage.
- * Its default behaviour is adding top & bottom margin. It is also a block element.
- * Syntax : `<p>` content `</p>`

Pre Tag <pre> :- (PreFormatted Tag)

- * This tag display the content exactly that written inside the pre tag including ~~the~~ the white space betⁿ the character & line break.
- * It display monospace font family. Syntax :- `<pre>` content `</pre>`

Div Tag <div> :-

- * It is a block element.
- * It is used for set a group of code & add section in the html document.
- * Syntax : `<div>` content `</div>`

Span Tag

- * It is an inline element, i.e used for formatting text in the same line content.
- * Syntax : `<span>` content `</span>`

Anchor Tag <a> :-

- * Anchor tags create hypertext on the webpage & that define a hyperlink by using href attribute.

* It is used for connect one page to another page for if you have click a hypertext, then you have to visit the destination page.

* Syntax: `<a href="hyperlink">content </a>`

href Attributes:- (Hyperlink Reference)

This attribute contain a destination page URL.

URL (Uniform Resource Locator):-

* A URL is the address of a unique resource in the internet.

* It is two types.

i) Absolute URL: A URL define by using https method.

ii) Relative URL: A URL define by using file name / folder name.

* This URL must be exist in the same parent folder where exist in the html file.

target Attribute:-

This attribute define where the destination page display such as display in current page & display another new current page.

Ex:- `<a href="index.html" target="_blank">index </a>`

-top
-parent
-self

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Image Tag:-

The image tag is used for display a image on web page by using src attribute.

Syntax:- ``

src Attribute / Source Attribute:-

It define the image path or URL.

Ex:- ``

``

alt Attribute / alternate attribute :-

* It define alternating text that display on the web page when the image path is wrong.

* Ex :- ``

Formatting Tag :-

* The Formatting tag are used for formatting the text without using CSS.

* The Formatting tags are;

`<b>` : It display text bold.

`<strong>` : It display text bold, but it define important text.

`<i>` : It display italic text.

`<em>` : It display italic text but it define emphasized text that means important text.

`<mark>` : It define a highlighted text & it display text by default background color yellow.

`<u>` : It display a underline text.

`<ins>` : It display a underline text, but it define inserted text.

`<s>` : It display line through on text.

`<del>` : It display line through on text but it define deleted text.

`<big>` : It display text font size bigger than the normal font size.

`<small>` : It display text font size smaller than the normal font size.

`<sub>` : It display the text half of below the normal line.

`<sup>` : It display the text half of above the normal line.

`<q>` : It add a quotation mark on the text.

List Tag :-

The list tag is used for display the content in list format like order list

& ~~and~~ unordered list.

OL Tag / Order list :- It is used to display the list content in orderwise like numeric & alphabetically.

Ex:- `<ol type="A">`

`<li>Monty</li>`

`<li>Khusi</li>`

`</ol>`

O/P
A. Monty
B. Khusi

`<ol type="1" start="5">`

`<li>Monty</li>`

`<li>Khusi</li>`

`</ol>`

O/P
5. Monty
6. Khusi

type Attribute :-

⇒ Type Attribute in OL tag define different order list marker point

Such as : a, A, i, I, 1

start Attribute :-

⇒ It define the initial point of list marker.

UL Tag / Unordered list :-

* It display the list content in unordered list format.

Such as : • bullet point, ○ circle, ■ square

Ex:- `<ul type="circle">`

`<li>Monty</li>`

`<li>Khusi</li>`

`</ul>`

O/P
○ Monty
○ Khusi

* It define different marker point in unordered list.

DL Tag <dl> / Description list :-

* It display a term & description the term in list format by using dt tag `<dt>` & dd tag `<dd>` / data description.

HTML Table :-

↳ The table is used for arranging the data, display in row and column format or in tabular wise.

↳ Table tag is used for create row & column layout in the webpage

<tr> : This tag create table row on a table.

<td> : td tag create table data cell which is display content in normal text.

<th> : This tag create heading data cell & display content, text bold & align center.

border attribute :- It create a border line to the table & create also border line each ^{table} ~~data~~ cell.

Cell spacing attribute : This attribute create or remove space betⁿ table each cell.

Cellpadding attribute :- It create space between table cell border & table cell content.

<table border="2" cellspacing="0" cellpadding="10">

<tr>

<th>Name </th>

<th>Age </th>

<th>College </th>

</tr>

<tr>

<td>Monty</td>

<td>15 </td>

<td>Cime </td>

</tr>

<tr>

<td>Khusi</td>

<td>16 </td>

<td>Cime</td>

</tr>

</table>

Name	Age	College
Monty	15	Cime
Khusi	16	Cime

rowspan

It merge two or more row cells by defining no. of rows.

Colspan

It merge two or more column cells by defining no. of columns.

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<thead>:-

This tag define table header content row.

It always display top of the row on the table.

<tbody>:-

This tag define body part content of table.

It display with in the header part & footer part of the table.

<tfoot>:-

This tag define footer content of table.

It display bottom of the row on the table.

<caption>:- Syntax:

This tag define a caption name of the table.

By default it display on the top of the table.

<Colgroup>:-

This tag content one or more ~~col~~ col tag & the col tag define each individual columns in the table.

```
<colgroup>
  <col1>
  <col2>
</colgroup>
```

Span Attribute:-

Span attribute in col tag specified one or more column at a time in specified number of columns.

Semantic Elements:-

- * The semantic elements describe the infoⁿ about their content
- * It make web pages for more accessible & navigate the content more efficient.
- * It SEO (Search Engine Optimization) friendly.
- * It helps to create logically structure document which is easier to read & maintain.
- * Such semantic elements are: `<table>`, `<header>`, `<body>`, `<footer>`...etc.

<Article>:-

- * It is used for content, that stand alone & that independently distribute. Such as blog post.

<aside>:-

It is used to create a side bar on the web page.

<header>:- It is used to add header part content on the web page like logo, navigation menu, login button.

<details>:-

This element define additional details that the user can hide & views.

<summary>:- It define a visible heading or a details element.

<figure> & <figcaption>

These are used to add an image to a webpage with a small description.

This description define by figcaption.

`<Figure>`

`<figcaption> Content </figcaption>`

`</Figure>`

<nav>:-

It is used to create navigation bar / nav menu on the webpage.

<main>:-

It define main content of the document.

The content inside the main tag should be unique.

<Footer>

It define footer content on the webpage such as contact details, Copyright info etc.

<Section>:-

It define a section or group of code in the html document.

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FORM:-

* The html elements create a form on the webpage.

* FORM is used for collect info from the user or site visitor.

* Syntax: `<Form action = " " > </Form>`

FORM Controls:-

input textfield

check box

clickable button

Select box

Radio box

File select box

Multiline input field

action attribute:-

* It define a URL where the form data will be processed after submit the form.

Method attribute:-

* This attribute create a http request after the form submit.

There are 3 values, `dialog` `get` `post`

i) Dialog: It can stop the http request when the form submit.

ii) get: It define http request & the form data visible in the URL after submit the form.

11) post :- It define http request & FORM data not visible in the URL after submit the FORM.

autocomplete attribute :-

This attribute control visible or hide the auto suggestion option when user focus on input text field

`<Form action=" " method=" " autocomplete`

novalidate attributes :-

^{stop}
It check the any type of validation in the FORM.

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Input element :-

* The input element create an input field where user enter info or data in the FORM.

* It is a inline block element.

`<input type=" " >`

Type attribute :-

⇒ Type attribute in the input field create different type input field like text field, select box, button, check box, radio box etc.

⇒ There are different types of value define different input ~~value~~ field.

⇒ Type attribute check a validation in the FORM when submit FORM.

`<input type="text">`

type="text"

It define text field that accept any type of character.

type="email"

It define input text field that accept only email address data.

type="number"

It define text field that allows to enter only numeric number.

type = "password" :-

It define text field it hide the data which is user entered in the text field.

type = "datetime"

It create text field that allow to enter date, month, year.

type = "search"

It define a search input field

type = "URL"

It define a text field that accept only URL

type = "tel"

It define a text field that entered a telephone number,

type = "hidden"

It define a hidden input field, that not visible on the web page.

type = "file"

It define a file select box, that allow to enter one or more file.

type = "date"

It define a select box that allow to select month, day & year

type = "month"

It define a select box that allow to select month & year.

type = "week"

It define a select box that allow to select week numbers & year.

type = "time"

It allow to select time that allow to select time.

type = "datetime-local" :-

It allow to select date with time at a time.

type = "color"

allow to select
It define a input field that, a color value.

type = "range"

It define a range bar that allow to select a number range betⁿ 2 number.

type = "checkbox"

It define a input checkbox button, i.e used for select one or more value.

type = "radio"

It define ~~radio~~ radio box that select one time a single name element.

type = "button"

It define a clickable button.

type = "reset"

It define a reset button that is used for clear all input field

Value.

type = "image"

It display a image on the webpage that is used for submitting the form.

type = "submit"

It define a submit button that is used for submitting the form.

Value attribute :-

This attribute set initial value in an input field.

name attribute :-

It define input element name.

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readonly attribute :-

This attribute define a readable input field that doesn't allow any changes. It work the following input type text, email, password, number, url, search, datetime etc

disable attribute :-

It define a disable input field that can't be any changes like ~~click~~ focus, enter value.

➤ Minlength attribute:-

This attribute allow to user enter minimum length of characters in the input field.

ii) maxlength attribute:-

This attribute allow to user enter max^m length of characters in the input field.

➤ These attributes work with input types text, email, datetime, url, tel, password etc.

Size attribute:-

It define visible width of the input field.

Its default value is "20".

Min & max attribute

These attribute set minimum value & max^m value in an input field.

Step attribute:-

This attribute set a legal number interval in an ~~ip~~ input field.

It work on range & number.

multiple attribute:-

This attribute allow to enter one or more file in an input field.

It work with input type email & input type file.

Placeholder attribute:-

* It is used for define^a short hint about an input element.

* That short hint means text i.e display behind the input text field.

* When we enter value in text field, then the placeholder can be hide

✗ the text field will be clear, then textholder place visible.

required attribute:-

This attribute define the user must be enter some value in the required input field before submitting the form. Otherwise

the form can't be submitted.

autofocus attribute:-

This attribute automatically focus in a input field when page load.

accept attribute:-

This attribute accept to select file that specify extension name.

`<input type="file" accept="image/jpeg">`
`accept = "image/png"`

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Label Element:-

It is used for focus an input element when some one click the label content.

Syntax: `<label for=" " ></label>`

for attribute:-

* The for attribute refers to input element for focus when for attribute value are same as the input id attribute value.

* One label element focus on an one input element

Select Option

* The select tag create a dropdown box for select option by using option tag.

* The option tag define option that can be selected

* By default the first option element can be selected in the dropdown list.

* We can define the pre selected option by using selected attribute in the option element.

`<select name=" " id=" " >`

`<option value=" " >-- Choose your state </option>`

`</select>`

Size Attributes:-

It define the visible number of option in the dropdown list.

<Text area> :-

It Create a multiline input field.

cols attribute:- It define visible width of the text area.

rows attribute:- The visible number of text line in the text area.

`<textarea name=" " id=" ">`

Fieldset Tag :-

The Fieldset element set a group of related data in the form.

Legend Element:- It define caption name of the fieldset element.

Datalist Element :-

The datalist element define a ~~pres~~ pre suggestion option for an input field.

The input list attribute value refers to a datalist element id attribute value.

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HTML Media

It means add some ~~med~~ multimedia content like image, audio, video or any type of document visible on the web page.

<audio>

The audio tag is used for display audio file on the webpage by using src attribute.

The src attribute in audio tag only accept relative url.

<video> :-

It used for display video on the web page.

The src attribute in a video tag only accept relative url.

Controls attribute :-

The controls attribute control to visible the audio & video on the web page & it also control play, pause, Forward, backward, Volume up, down etc.

autoplay attribute :-

This attribute automatically play the video and audio when page load.

muted attribute :-

This attribute mute the volume automatically when page load.

loop attribute :-

This attribute repeatedly play the audio & video on the webpage.

iframe Tag :-

iframe stands for inline frame.

* This tag is used for display an another document on the current document.

* We can use iframe tag, we can display any type of document on the webpage.

<Marquee> :-

This tag add scrolling effect on the webpage by using direction

`<marquee behaviour=" " direction=" " > </marquee>`

<Center> :-

This tag is use for align the content or element to the center of the webpage.